A063 – Hicks/Hill Euler Separatrix Falsifier

Scientific report – v0.1.1

Scope: Exact-field analytic reference, interface convergence, and numerically closed toroidal streamline candidate.
Evidence: Python/SciPy FP64, analytical/synthetic control only. **Not** canonical v1.0.7 native certification.
Blind status: no target particle labels, canonical masses or SST parameter fitting.

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1 Executive verdict

A total of **10 PASS**, **3 UNRESOLVED**, and **0 FAIL** gates were recorded. The vortex-field reference is consistent with its prescribed accuracy gates; the original global topological/stability gate `H7`, the new trefoil/tube gate `H7B`, and independent native certification `H8` remain `UNRESOLVED`. A `PASS` of `H7A` only establishes a *numerically closed T(3,4) streamline candidate*.

This report reads the additive v0.1.1 result JSONs. No older A063 v0.1.0 verdict is changed or migrated. Calculations are dimensionless with representative sphere radius $a = 1$, density $\rho = 1$ and predefined amplitude $A = 1$.

2 Exact Euler controls

For cylindrical coordinates (s, ϕ, z) and the sphere $r^2 = s^2 + z^2 \le a^2$, the streamfunction and velocity in a sphere-comoving frame are

$$J(x) = \frac{\sin x}{x} - \cos x, \tag{1}$$

$$\psi_\lambda(r,\theta) = A\left[J(\lambda r/a) - \frac{r^2}{a^2}J(\lambda)\right]\sin^2\theta, \tag{2}$$

$$(u_s, u_\phi, u_z) = \left(-\frac{\psi_z}{s}, \frac{\lambda\psi}{as}, \frac{\psi_s}{s}\right). \tag{3}$$

At $\lambda = 0$ a separate cancellation-safe Hill limiting field is used. The special families satisfy

$$J(\lambda_1) = 0 \quad \text{(Beltrami family)}, \tag{4}$$

$$3J(\lambda_2) - \lambda_2 \sin\lambda_2 = 0 \quad \text{(zero-translation family)}. \tag{5}$$

With the local velocity gradient and Euler pressure obtained from the same analytic field, the interior reference checks

$$\nabla\cdot\mathbf{u} = 0, \qquad (\mathbf{u}\cdot\nabla)\mathbf{u} + \frac{\nabla p}{\rho} = 0, \qquad \nabla\times\mathbf{u} = \frac{\lambda_1}{a}\mathbf{u} \text{ for the Beltrami solution.} \tag{6}$$

Methodological correction from v0.1.0: the residual maxima are now computed *separately* for Hill, first Beltrami, and first zero-translation Hicks; the former H3 pooled Euler/divergence maxima from distinct families. Four finite-difference steps, $h/a \in \{4, 2, 1, 0.5\} \times 10^{-4}$, are compared for each family.

Reference	Max Euler residual (finest)	Max $\lvert\nabla\cdot\mathbf{u}\rvert$ (finest)
Hill	9.22e-09	4.36e-12
Beltrami	8.87e-07	3.26e-08
Zero translation	4.26e-06	1.08e-07

The finest relative Euler error reduction ratios (previous step / finest step) exceed 2.5 for all three cases. These are finite-sample diagnostics, not global validated error bounds.

3 Full-latitude material boundary and continuity

The spherical separatrix is tested at 121 polar angles in each of three fields. The analytic no-flux condition $\mathbf{u}\cdot\mathbf{n} = 0$ is evaluated on the sphere. Interior/exterior jumps in tangential velocity and pressure are measured at $r = a(1 \pm \epsilon)$ for $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}\}$.

Field	$\Delta u(10^{-3})$	$\Delta u(10^{-5})$	$\Delta p(10^{-5})$
Hill	0.005988	5.995e-05	4.5e-05
Hicks_Beltrami_1	0.02157	0.0002157	0.0003848
Hicks_static_1	0.03169	0.0003169	3.425e-13

The jumps decay with radial offset for all fields. The latitude scan excludes only small caps $\theta < 0.06$ and $\pi - \theta < 0.06$, which remain outside this finite-sample claim. A smooth-interface test does not prove dynamical self-confinement.

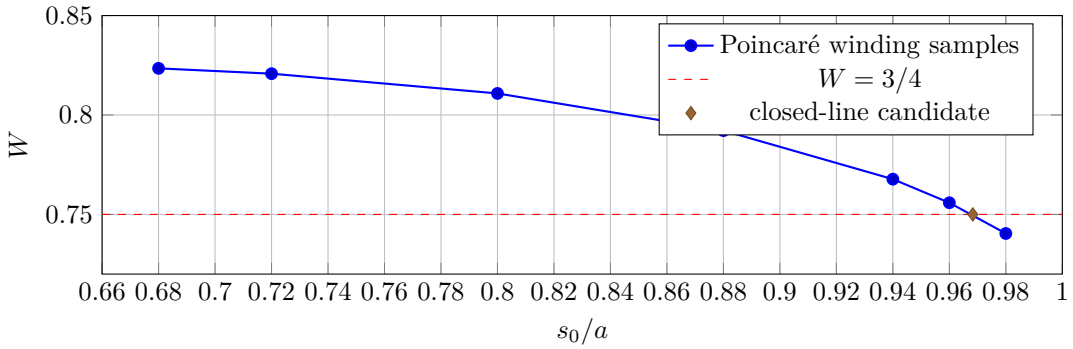
4 First rational-winding torus-line test

On the first Beltrami family we integrate the Poincaré return map of the meridional plane and define

$$W(s_0) = \frac{\Delta\phi}{2\pi} \quad \text{per complete meridional circulation.} \tag{7}$$

The target $W = 3/4$ is bracketed by two independent sampled curves, then solved using Brent's method. Continuous three-dimensional integration over four meridional periods independently tests closing at a primitive $(p, q) = (3, 4)$ winding:

$$s_0/a = 0.968309830174, \tag{8}$$
$$W(s_0) = 0.750000000000, \tag{9}$$
$$(N_{\text{azimuth}}, N_{\text{meridional}}) = (3.000000000, -4.000000000), \tag{10}$$
$$\|\mathbf{x}(4T) - \mathbf{x}(0)\|/a = 9.1832e - 10. \tag{11}$$



Allowed claim: a numerical $T(3, 4)$ *streamline candidate* on an axisymmetric invariant torus, with monotone meridional phase and positive minimum radius. This does **not** independently certify a knot isotopy (e.g., with an Alexander polynomial), and does not show a finite-radius vortex *tube*, an electron, or nonlinear Euler stability. It also does not globally exclude trefoil $T(2, 3)$ from higher modes or other parameter ranges.

5 Analytic bridges for the existing SST falsifiers

A046 pressure–vorticity: tests $\nabla^2 p = -\rho\, \partial_i u_j \partial_j u_i$ in the incompressible Euler field, without interpreting p as gravitation. Maximum sampled pressure–Poisson absolute residual: 0.00028.

A057 finite-volume reference: computes total laboratory-frame kinetic energy from interior volume quadrature plus the exact exterior dipole energy. Uniform far-field kinetic energy is not integrated in the translating frame. No constant-circulation normalization or physical mass-ratio inference is performed.

Field	Interior E	Exterior E	Total E
Hill	3.440792	1.047198	4.48799
Beltrami_1	98.49107	8.953734	107.4448
Static_1	152.0015	3.671525e-31	152.0015

6 Gate ledger and remaining qualifications

Gate	Status	Registered question
H0	PASS	Versioned source/protocol bytes match release seal?
H1	PASS	Do analytical roots satisfy registered equations?
H2	PASS	Does Hill zero-swirl control converge under finite-difference refinement?
H3	PASS	Does first Hicks Beltrami field satisfy curl(u)=k*u, Euler and divergence under refinement?
H4	PASS	Does lambda2 first Hicks mode have zero translation and a convergent stationary Euler field?
H5	PASS	Is the spherical boundary a material surface in all three exact fields?
H6	PASS	Do full-latitude interior/exterior pressure and velocity jumps converge to zero?
H7	UNRESOLVED	Are torus-knot accessibility and finite-core vortex-tube stability both certified?
H7A	PASS	Can a first-Beltrami streamline realize a closed rational 3/4 torus-winding candidate?
H7B	UNRESOLVED	Is trefoil T(2,3) accessible as a finite-core stable tube?
H8	UNRESOLVED	Has independent C++ FP64 certification run through v1.0.7 frozen protocol?
H9	PASS	Does exact pressure satisfy incompressible pressure-Poisson closure?
H10	PASS	Does finite laboratory-frame kinetic-energy quadrature converge?

Immediate next work: independent topological invariant from sampled closed curve; global interval winding accessibility bounds for the trefoil; perturbed finite-core Euler-vortex-tube evolution and convergence; separate C++ FP64 native lane with the unmodified v1.0.7 instance protocol if certification is requested. No legacy falsifier framework migrations are necessary to render PDF reports.

Source and provenance

Primary analytic source: W. M. Hicks (1899), *Researches in Vortex Motion. Part III: On Spiral or Gyrostatic Vortex Aggregates*, *Phil. Trans. R. Soc. A* **192**, 33–99, DOI: 10.1098/rsta.1899.0002. Classic reference: M. J. M. Hill (1894), *On a Spherical Vortex*, *Phil. Trans. R. Soc. A* **185**, 213–245, DOI: 10.1098/rsta.1894.0006. This release is an analytical Python control campaign, with no independent observational support for SST particle physics.
Run protocol SHA-256: `838a73226435889fe3cfee63053ee12d6a4a06d7cde67bb74a7ad02296038b99`.
Ledger SHA-256: `5283d44e0cf4a1ae4f514c6477169fac5e4be50e191daa5a637b07b9850740c4`.
Run timestamp (UTC): 2026-10-13T14:41:39.972373+00:00.